

THE COST OF RANDOM GROUPS WITH DEFICIENCY AT LEAST 2

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ABSTRACT. Write abstract

1. INTRODUCTION

Let G be a countable group and let X be a probability space on which G acts by measurable transformations, endowed with a G -invariant probability measure μ . One says that G acts by probability-measure preserving (p.m.p.) transformations on (X, μ) .

The *cost* of a countable group G is $\frac{1}{2} \inf_{\mathcal{G}} \{\mathbb{E}_{\mathcal{G}}(\deg(e_G))\}$, where the infimum ranges over all G -invariant connected graphs $\mathcal{G}$ with vertex set G , and where $\mathbb{E}_{\mathcal{G}}(\deg(e_G))$ is the expected number of edges in $\mathcal{G}$ that have e_G as an endpoint.

The sofic dimension of a p.m.p. action...

Definition 1.1. Let G be a finitely generated residually finite group.

- The *absolute rank gradient* of G is

$$\mathbf{rg}_{\text{abs}}(G) := \inf \left\{ \frac{d(H) - 1}{[G : H]} : H \leq G \text{ with } [G : H] < \infty \right\}, \quad (1)$$

where $d(H)$ denotes the smallest size of a generating set for the group H .

- The *absolute subgroup growth gradient* of G is

$$\mathbf{sgg}_{\text{abs}}(G) := \inf \left\{ \frac{1}{[G : H]} \limsup_{d \rightarrow \infty} \frac{\log(s_H(d))}{d \log(d)} : H \leq G \text{ with } [G : H] < \infty \right\}, \quad (2)$$

where $s_H(d) := |\{K \leq H : [H : K] \leq d\}|$, $d \geq 1$, is the subgroup growth function of the group H .

The inequalities

$$\beta_1^{(2)}(G) \leq \text{cost}(G) - 1 \leq \mathbf{rg}_{\text{abs}}(G), \quad (3)$$

and

$$\mathbf{sgg}_{\text{abs}}(G) \leq s(G, \hat{G}) - 1 \leq \mathbf{rg}_{\text{abs}}(G) \quad (4)$$

always hold. There are no known examples of groups where they are strict. Our main theorem asserts that on a random group of deficiency at least 2, these inequalities are equalities with probability tending to 1.

Theorem 1.2. *Let G be a random group in the few relators model with $n \geq 3$ generators and $m \geq 0$ relations. Suppose that $m \leq n - 2$. Then with asymptotic probability 1 we have*

$$n - m - 1 = \text{cost}(G) - 1 = \mathbf{rg}_{\text{abs}}(G) = \mathbf{sgg}_{\text{abs}}(G) = s(G, \hat{G}) - 1.$$

2. PRELIMINARIES

2.1. Cost and rank gradient.

2.2. Sofic dimension and subgroup growth gradient.

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2.3. The few relators model for a random group.

Definition 2.1. Let $n > m \geq 1$. Let F_n be the free group of rank n with free basis $\{x_1, \dots, x_n\}$ and let $(r_1, \dots, r_m)$ be an m -tuple of cyclically reduced words on F_n . Let $\phi : F_n \rightarrow \mathbb{Z}$ be a homomorphism. The pair $((r_1, \dots, r_m), \phi)$ satisfies the *minimum condition* if, possibly after renaming the generators $\{x_1, \dots, x_n\}$, for each $j = 1, \dots, m$ it holds that $\phi(r_j) = 0$ and that the lower section $L_\phi(r_j)$ consists of

- either exactly one vertex, whose two adjacent edges are labeled by $x_i^{\pm 1}$ and $x_n^{\pm 1}$,
or
- exactly one edge, labeled by $x_i^{\pm 1}$, whose adjacent edges are labeled by $x_n^{\pm 1}$.

The pair $((r_1, \dots, r_m), \phi)$ satisfies a *minimum condition* if there exists $\phi : F_n \rightarrow \mathbb{Z}$ such that, in addition to the above, satisfies $\phi(x_i) \geq 0$ for each $1 \leq i < n$ and $\phi(x_n) < 0$.

Proposition 2.2 ([KKW22, Proposition 5.3]). *A random group in the few relators model with $n \geq 3$ generators and $m \leq n - 2$ relators satisfies a standard minimum condition with asymptotic probability 1.*

3. AN UPPER BOUND FOR THE ABSOLUTE RANK GRADIENT

Theorem 3.1. *Let $G = \langle x_1, \dots, x_n \mid r_1, \dots, r_m \rangle$ be a finitely presented infinite group with $n > m + 1$. Suppose that G is residually finite and that the m -tuple $(r_1, \dots, r_m)$ satisfies a standard minimum condition. Then there exists a chain $(G_k)_{k \geq 0}$ of finite index normal subgroups of G with vanishing intersection such that $\mathbf{rg}(G, (G_k)_{k \geq 0}) = \mathbf{rg}_{\text{abs}}(G)$ and*

$$\text{cost}(G, \varprojlim G_k) \leq n - m.$$

4. A LOWER BOUND FOR THE ABSOLUTE SUBGROUP GROWTH GRADIENT

Theorem 4.1. *Let $G = \langle x_1, \dots, x_n \mid r_1, \dots, r_m \rangle$ be a finitely presented infinite group with $n > m + 1$. Suppose that G is residually finite and that the m -tuple $(r_1, \dots, r_m)$ satisfies a standard minimum condition. Then there exists a chain $(G_k)_{k \geq 0}$ of finite index normal subgroups of G with vanishing intersection such that $\mathbf{srg}(G, (G_k)_{k \geq 0}) = \mathbf{srg}_{\text{abs}}(G)$ and*

$$s(G, \varprojlim G_k) \geq n - m.$$

5. THE PROOF OF THE MAIN THEOREM

REFERENCES

[KKW22] Dawid Kielak, Robert Kropholler, and Gareth Wilkes. ℓ^2 -Betti numbers and coherence of random groups. *J. Lond. Math. Soc. (2)*, 106(1):425–445, 2022. [Cited on page 2.]

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